

BT302

Assignment 2, Fall 24

Total Marks: 10

Due date: 6th-Dec-2024

Instructions:

- Make sure that you upload the solution before due date. No assignment will be accepted through e-mail after the due date.

Formatting guidelines

- Use the font style "Times New Roman" and font size "12".
- Use black and blue font colours only.

Solution guidelines

- To solve this assignment, you should have good command over lectures 11 to 14
- This is not a group assignment; it is an individual assignment so be careful and avoid copying others' work
- Give the answer according to question only and avoid irrelevant details.

Please note that your assignment will not be graded if:

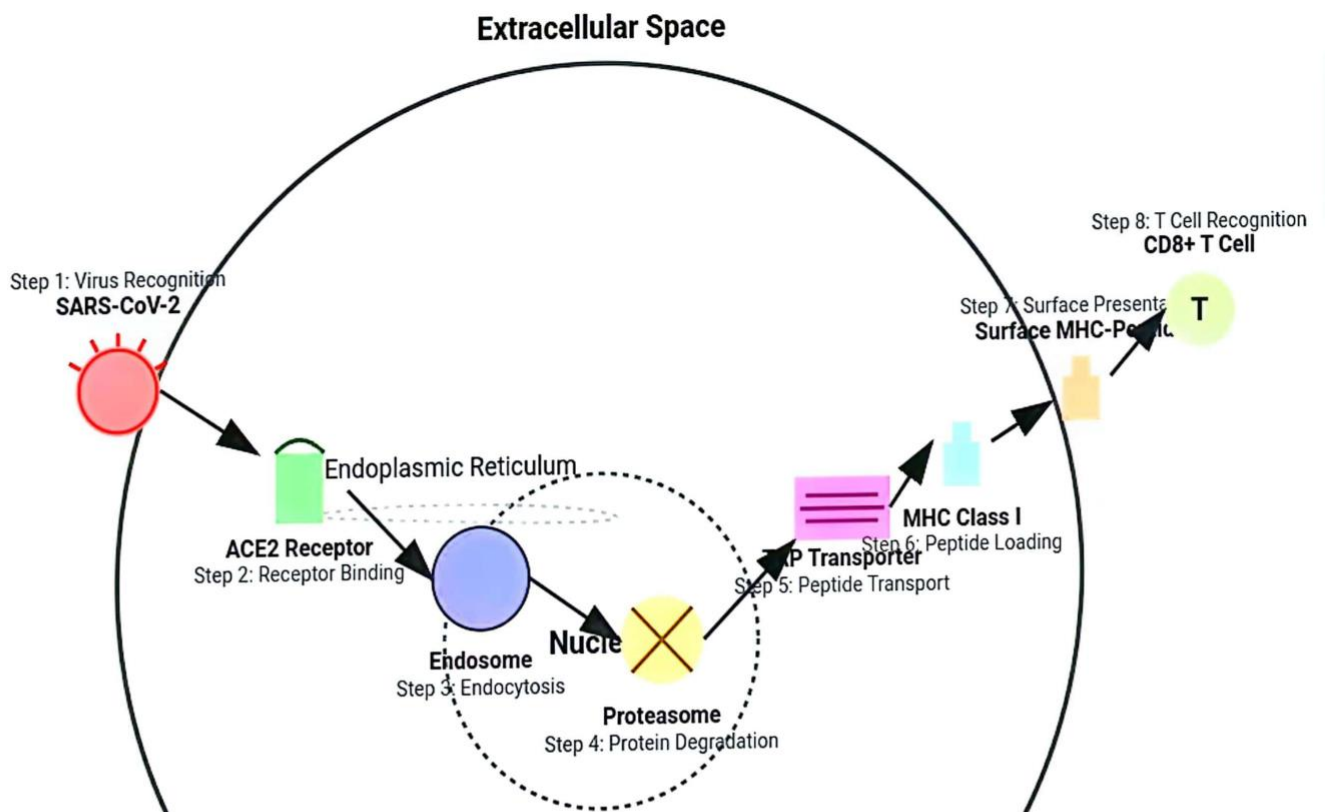
- It is submitted after due date
 - Cheating or copying of assignment is strictly prohibited. The cheated or copied assignment will be marked '**Zero**'.
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Question:

Create a visual representation of how SARS-CoV-2 (Coronavirus) antigens are processed and presented through the MHC. The diagram should be specific to SARS-Cov-2 and not any other virus.

- Text-only answers, hand-drawn diagrams, AI-generated diagrams and diagrams taken from the internet will receive a zero.
- Use insert image to paste it in solution

SARS-CoV-2 Antigen Processing and Presentation via MHC Class I (Tentative Illustration)



Detailed Process Steps:

1. SARS-CoV-2 virus approaches the cell surface
2. Viral spike proteins bind to ACE2 receptors on cell membrane
3. Virus enters cell through receptor-mediated endocytosis
4. Viral proteins are degraded into peptide fragments by proteasome
5. Peptides are transported into ER lumen via TAP complex
6. Peptides are loaded onto newly synthesized MHC Class I molecules
7. MHC-peptide complexes are transported to cell surface
8. CD8+ T cells recognize viral peptides presented by MHC-I

1. Viral Entry

- **Receptor Binding:** SARS-CoV-2 binds to the ACE2 receptor on the host cell surface.
- **Entry Mechanism:** The virus enters the cell through endocytosis or membrane fusion.

2. Viral Protein Synthesis

- **RNA Release:** The viral RNA genome is released into the cytoplasm.
- **Protein Production:** Host ribosomes translate viral RNA into structural and non-structural proteins.

3. Proteasomal Processing

- **Degradation:** Viral proteins are degraded by the proteasome into short peptides (8–10 amino acids) suitable for MHC class I binding.

4. Peptide Transport to the ER

- **TAP Transporter:** Peptides are transported into the endoplasmic reticulum (ER) via the Transporter Associated with Antigen Processing (TAP).

5. MHC Class I Loading

- **Peptide Binding:** Peptides bind to MHC class I molecules in the ER.
- **Complex Stabilization:** The binding stabilizes the MHC class I-peptide complex for transport.

6. Transport to the Cell Surface

- **Golgi Apparatus:** The MHC class I-peptide complex passes through the Golgi apparatus.
- **Surface Presentation:** The complex is presented on the host cell surface.

7. Recognition by CD8+ T Cells

- **Immune Activation:** CD8+ cytotoxic T lymphocytes (CTLs) recognize the viral peptides through their T cell receptors (TCRs).
- **Response:** CTLs target and destroy infected cells, limiting viral replication.

Key Components

- **ACE2 Receptor:** Facilitates viral entry.
- **Proteasome:** Processes viral proteins into peptides.
- **TAP Transporter:** Delivers peptides to the ER.
- **MHC Class I Molecules:** Present peptides on the cell surface.
- **CD8+ T Cells:** Recognize the antigen and initiate immune responses

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